

TECHNICAL BULLETIN

DIXIE CHEMICAL

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EPOXY CURING AGENT 608 (ECA 608)

Typical Properties

Property	Value	Test Method
Molecular Weight (average)	172 - 174	
Specific gravity @ 25°C, g/cc	1.155 - 1.159	D-102 (ASTM D 4052)
Appearance	Green Liquid	D-104 (Visual)
Viscosity (Brookfield at 25 °C), cP	50 - 200	D-150

Applications

ECA 608 is a low viscosity, non-crystallizing liquid anhydride epoxy curing agent blend. For convenience, it is pre-catalyzed with latent proprietary accelerators that impart high reactivity at process temperatures, while maintaining excellent pot life. It is designed to cure efficiently in processes such as pultrusion where residence time at curing temperature is limited. ECA 608 has the following characteristics:

- **Easy Handling:** Low viscosity liquid gives low viscosity mixes with various epoxy resins
- **Convenience:** Eliminates the need to add small amounts of catalyst
- **Consistency:** Prevents potential batch-to-batch differences caused by errors in measuring and properly dispersing small amounts of catalyst

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- **Low reactivity at room temperature:** Provides long working times and pot lives for mixes with epoxy resins. Formulations of ECA 608 and liquid epoxies have a long pot life/working time in excess of 24 hours.
- **High reactivity at cure temperature:** Gives excellent cure response at cure temperatures, resulting in high crosslinking and relatively high Tg.

For more information on the use of anhydrides like ECA 608 as epoxy curing agents, please consult the Technical Bulletin, FORMULATING ANHYDRIDE-CURED EPOXY SYSTEMS, available from Dixie Chemical Company.

Formulation Example

ECA 608 was formulated with a variety of epoxy resins. Viscosity was measured at 25°C, and gel time was measured at 100°C. Samples were cured for 1 hour at 120°C, followed by a post-cure of 1 hour at 220°C, and glass transition temperatures were measured using a differential scanning calorimeter. The following table summarizes these results:

Epoxy type	ECA 608, phr	Viscosity at 25 °C, cP	Gel Time ¹ , min	Cured Tg ² , °C
Standard BPA Liquid Epoxy	89	750	20	152
Low Viscosity BPA Liquid	92	580	19	151
Cycloaliphatic Epoxy	122	120	16	231
Epoxy Phenol Novolac	96	680	19	150
Epoxy BPA Novolac	87	2330	19	161
BPF Liquid Epoxy	96	400	19	138

1. Shyodu Hot Pot (100g at 100°C)

2. Cured 1 hr at 120°C and 1 hr at 220°C. Analyzed in DSC: 40°C/min to 250°C

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Comparative Study

Different pre-catalyzed anhydrides were formulated with a standard BPA liquid epoxy resin. The following table summarizes these results:

Anhydride type	Viscosity at 25 °C, cP	Onset Temperature, °C	Gel Time ¹ , min	Cured Tg ² , °C
ECA 608	750	91	20	152
ECA 607	850	107	24	150
ECA 609	810	90	16	150
ECA 100NC with 2% BDMA		115	18.6	140
ECA 100KA	1560	120	33	144

1. Shyodu Hot Pot (100g at 100°C)

2. Cured 1 hr at 120°C and 1 hr at 220°C. Analyzed in DSC: 40°C/min to 250°C

Handling & Storage

ECA 608 will react with water to form diacids. This is normally undesirable, so ECA 608 should be stored in such a way that it is carefully protected from moisture contamination.

For more details on the design of bulk storage for ECA 608, consult the Dixie Chemical Company brochure "Epoxy Curing Agent Storage Requirements."

Health Hazards

Read and understand the relevant Safety Data Sheets (SDS) for all products before use. These anhydrides are primary skin and eye irritants. Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, follow the procedures outlined in the SDS.

The data contained herein are furnished for information only and are believed to be reliable. This information is provided only as guidance and is not to be considered a warranty or quality specification. Dixie Chemical Company, Inc. makes no warranties (expressed or implied) as to its accuracy and assumes no liability in connection with any use of this product.

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Generally, these procedures include immediately flushing the affected skin or eyes with copious amounts of water for at least 15 minutes. In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

Follow the recommendations in the SDS for personal protective equipment when handling these materials. At a minimum, these procedures typically include protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

ECA 608 is available from Dixie Chemical Company, Inc.'s plant in Pasadena, Texas. Contact your Dixie representative for details.